Lecture III - Caching

Programming: Everyday Decision-Making Algorithms

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Learning Objectives

By the end of this lecture, you will be able to:

- Explain the fundamental concepts of caching and its importance
- Compare different cache replacement strategies
- Identify caching principles in everyday life
- Apply caching concepts to personal productivity
- Understand the relationship between caching and attention management

Introduction

A Quick Question First

Question: **How many of you have a messy desktop right now?**

Raise your hand!

Today's lecture will explain why that matters more than you think...

Let’s approach the topic using an everyday decision

- We have a problem: Our cupboard.
- It’s time to put things in order.

Question: What could we do?

- Better organization
- Clearing out things we no longer need
- Now we have two problems:
  - Storing?
  - Clearing out?

Two Approaches to Storage Problems

- **Better Organization:**
  - Subdivide storage
  - Efficient sorting
- **More Space:**
  - Increase capacity
- Question: Which approach is better?

Even best organization has limits

- Organization helps, but takes time.
- More space helps, but has limits.
- **Every storage has a finite capacity.**

Question: What do we do, when the storage is full?

We could increase the capacity

But...

- Increasing capacity = costly
- **Trade-off:** Size vs. Speed
- Larger = slower to search
- Sooner or later: Still fills up

Question: What else faces this problem?

- Cupboards
- Computers
- Email inbox
- Smartphones
- Warehouses
- **Our brain?!**

Question: Impact of full storage?

- Access speed drops
- Processing time up
- Performance down

Clearing out

Why Clearing Out Matters

- True for cupboards, computers, brains...
- **But what stays and what goes?**

Learning from Computer Science

The evolution of computer memory

- 1950s: Computer science faced the same problem
- Processors got faster (Moore's Law)
- Memory demands grew
- But memory speed couldn't keep up

→ **The Memory Wall**

The Bottleneck

- **Modern CPUs:** Billions of ops/second
- **Problem:** Data isn't available fast enough
- Question: What's the point of a fast CPU if it has to wait for slow memory?

→ **Von Neumann Bottleneck**

Cache

Cache: The Solution

A hierarchical memory system

- **L1 Cache:** Tiny but ultra-fast (64-256 KB)
- **L2/L3 Cache:** Larger, still fast (MB range)
- **RAM:** Main workspace (8-32 GB)
- **Storage:** Huge but slow (256 GB - 2 TB)

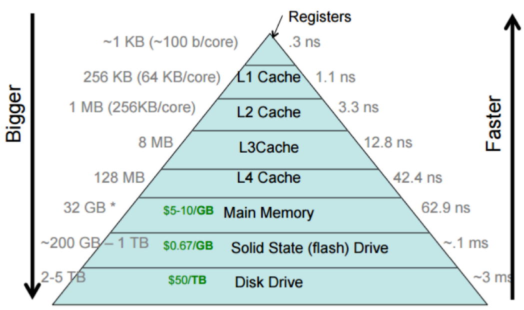
Like a library...

The Library Principle

- **Library storage** (5 million books, Mass Storage)
- **Subject locations** (100K books, RAM)
- **Your desk** (5 borrowed books, L2)
- **Short-term memory** (current page, L1)

Closer = Faster = Smaller

The Trade-off: Size and Speed



Registers are 10 million times faster than the hard drive!

Why can't we just make everything as fast as L1 cache?

Why the Trade-off?

Why not make everything ultra-fast?

- **Physical limits:** Larger caches sit further from CPU, signals take longer to travel, and dense fast memory generates extreme heat that must be dissipated
- **Economic limits:** SRAM costs $50,000/GB vs. HDDs at $0.02/GB

- **Solution:** Multiple cache levels, each optimized for different needs

The nessecity of clearing out

- L1 and L2 cache only contain most necessary data.
- The same should apply to your desk.
- Therefore, both must be cleared regularly.

Clearing out Strategies

Question: Eviction Strategies?

- Random
- First-In, First-Out (FIFO)
- Least Frequently Used (LFU)
- Least Recently Used (LRU)

How to clear up?

- **Optimal: Clairvoyance**
  - Keep what you'll need
  - Remove what you won't
- Question: What's the problem?

→ Requires knowledge of the future!

Realistic Strategies

- **Least Recently Used (LRU)** is the dominant strategy.
- Evicts the least recently accessed item from the cache when space is needed.
- Leads to much better performance on average than, for example, random eviction.
- Question: Why do you think least recently used is the better strategy?

Why LRU Works: The Principle of Temporal Locality

- **Temporal Locality:** Recent use → Likely need again soon
- **Examples:**
  - Books on your desk
  - Apps opened today
  - People texted this morning
- **Performance:** 80-90% hit rate (vs. 50-60% random)

Recent past predicts near future

Managerial and personal insights:

- **Let go of unused things** → LRU principle
- **Keep things where used** → Spatial locality
- **Result:** Significant productivity increase

Marie Kondo = LRU for physical objects!

Spatial Locality

Question: Can you think of examples where spatial locality is applied in your daily life?

Mathematically optimal

Productivity

Our Brains are Caches

- We've learned how computers manage limited cache space
- **But why does this matter for humans?**
- Your brain works remarkably similar to a computer cache:
  - Limited capacity for active information
  - Fast access to recently used information
  - Must constantly decide what to keep and what to forget

Cache Vulnerabilities

- **Denial-of-Service (DoS) attacks** exploit cache limitations:
  - **Cache Flooding:** Overload with excessive requests
  - **Cache Poisoning:** Insert malicious data to evict important information
- These attacks overload a system with excessive requests or data.
- Causing it to slow down or crash.
- The system is forced to evict important data.

Productivity Killers

- **Overload** (too much information, cache capacity exceeded)
- **Exhaustion** (too long without "cache clearing")
- **Context switching** (interruption of "flow", ~23 minutes to get back on track)
- **Distraction** (Cache Flooding: constant notifications, social media)
- **Misinformation** (Cache Poisoning: fake news, misleading information)

The Attention Crisis

- **Technology & Social Media:** Designed to capture attention (Cache Flooding)
- **Constant Interruptions:** Notifications = Forced cache evictions
- **Result:** Reduced creativity, critical thinking, and productivity

Reflection: How many times did you check your phone during this lecture?

The Solution: Manage Your Mental Cache

- Limit screen time and practice “monotasking” (Reduce cache thrashing).
- Prioritize sleep, nutrition, and mindful habits (Maintain cache performance).
- Create protected time for deep work (Prevent cache flooding).
- Be selective about what enters your attention (Smart cache management).

Mitigation

Mitigation

- Distraction can hardly be avoided in today’s world but can be mitigated.
- This is particularly important for managers.
- This lecture is designed to raise your awareness of what you can do to keep your brain working efficiently.
- **Want to dive deeper? Read “Stolen Focus” by Johann Hari**

Awareness: Three Levels of Attention

- **Spotlight** – Immediate goals – Focus (Your L1 Cache: What are you working on RIGHT NOW?)
- **Starlight** – Medium-term goals – Wishes (Your L2 Cache: What matters this week/month?)
- **Daylight** – Long-term goals – Values (Your Storage: What defines your life direction?)

Practical Strategies

- **Prioritization:** What deserves to be in your mental cache?
- **Structure (Schedule):** Time-boxing prevents cache overflow
- **Breaks:** Regular “cache clearing” prevents exhaustion
- **Enable flow:** Dedicated workspace, manage notifications, clear communication
- **Meditation & exercise:** Maintenance routines for optimal cache performance

Cache Hierarchy for Workspace

- **Hot Cache (On your desk):** Only items for current tasks (today)
- **Warm Cache (Nearby shelf/drawer):** This week’s projects (this week)
- **Cold Storage (Archive/closet):** Everything else

→ **The key principle:** If you haven’t used it recently, move it away!

Self-Assessment

Question: **Right now:**

1. How many browser tabs do you have open? ___
2. How many notifications did you receive in the last hour? ___
3. How many unread emails in your inbox? ___

Do you see any risks of cache flooding?

Key Takeaways

Key Takeaways: The Concepts

- **Caching is universal:** It applies to computers, organizations, and human cognition
- **The fundamental trade-off:** Fast storage is small, large storage is slow
- **LRU works because of temporal locality:** Recent past predicts near future

Key Takeaways: Your Life

- **Your brain is a cache:** Limited capacity, vulnerable to flooding and poisoning
- **Protect your attention:** It's your scarcest and most valuable resource
- **Apply caching principles to life:** Keep what you use, remove what you don't

Literature

Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[1]
- Ferguson, T.S. (1989) 'Who solved the secretary problem?', Statistical Science, 4(3). doi:10.1214/ss/1177012493.

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Here
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

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> **Note**
>
> Think Python is a great book to start with. It's available online for free. Schrödinger Programmiert Python is a great alternative for German students, as it is a very playful introduction to programming with lots of examples.

[1] The main inspiration for this lecture. We have read it and discussed it in depth, always wanting to translate it into a course.

More Literature

For more interesting literature, take a look at the literature list of this course.